

Air University



SSDD (Lab)

Year 2026

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Academic Year: 2024 – 2028

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BS – Cyber Security – Fall – 24 (A)

Date of Submission: February 17, 2026

Lab Task # 04**System Scenario: Online Shopping System**

An online shopping system allows:

- Customers to register/login
- Browse products
- Add items to cart
- Make payment
- Track orders

Admin can:

- Add / update products
- Manage orders
- View reports

PART 1 – Use Case Modeling (Functional View)**Task 1: Identify Actors**

Students must identify:

- Customer
- Admin

Task 2: Identify Main Use Cases

Examples students should include:

For Customer:

- Register
- Login
- Search Product
- Add to Cart
- Checkout
- Making Payment
- Track Order

For Admin:

- Manage Products
- Manage Orders
- Generate Reports

Task 3: Draw Use Case Diagram**Students must:**

- Use UML notation
- Show relationships:
 - <>
 - <>
 - Generalization (if applicable)

Example:

- Checkout <> Login
- Checkout <> Make Payment

PART 2 – Misuse Case Modeling (Security View)

Now, students think like attackers

Task 4: Identify Misuse Actors**Examples:**

- Hacker
- Fraudster
- Malicious User

Task 5: Identify Misuse Cases**Examples:**

- Steal user credentials
- Perform SQL Injection
- Make a fake payment confirmation
- Access the admin panel without authorization
- Modify product price

Task 6: Link Misuse Cases to Use Cases**Students must:**

- Show which normal use case is being threatened
- Use “threatens” relationship

Example:

- “SQL Injection” threatens “Login.”
- “Fake Payment” threatens “Make Payment.”

PART 3 – Security Controls (Mitigation)

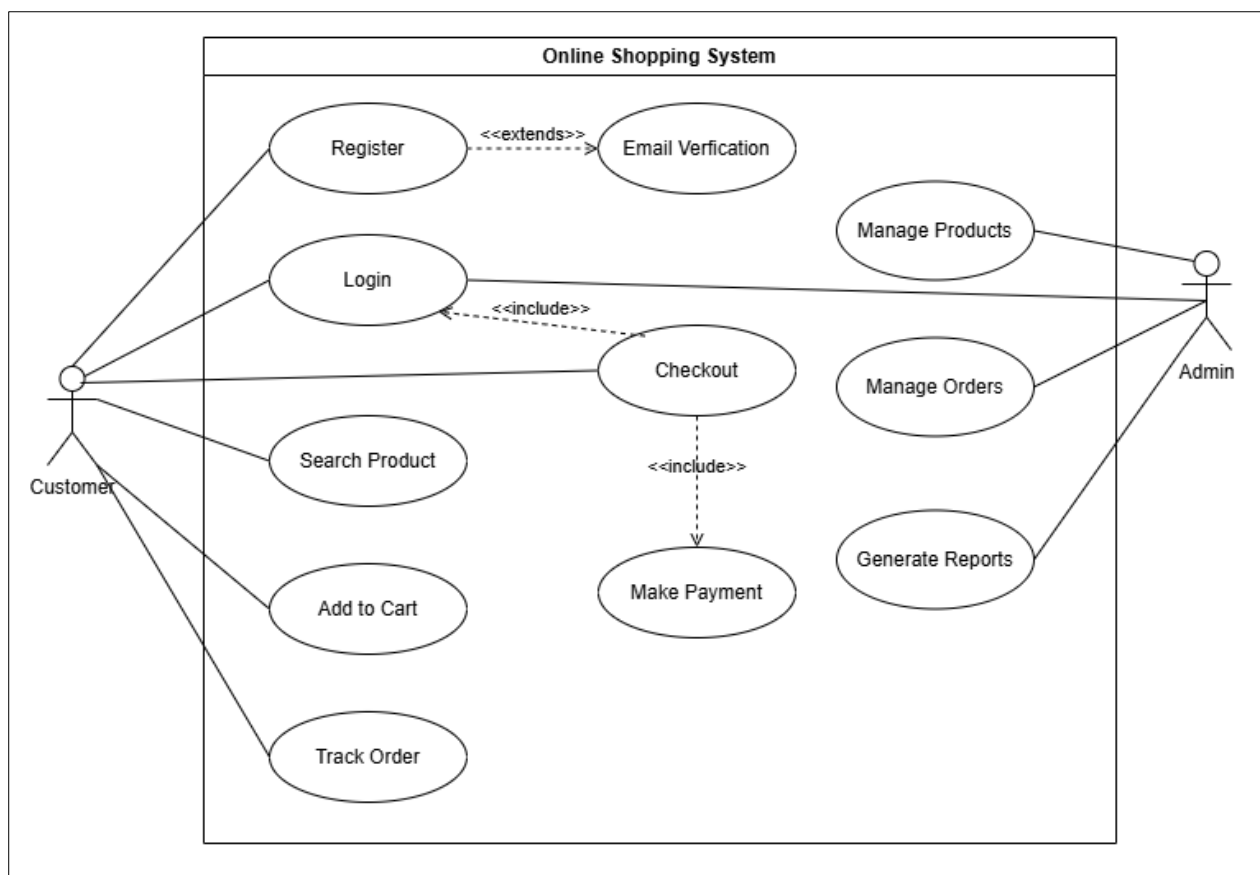
Students must propose security controls:

Credential theft, MFA, Encryption

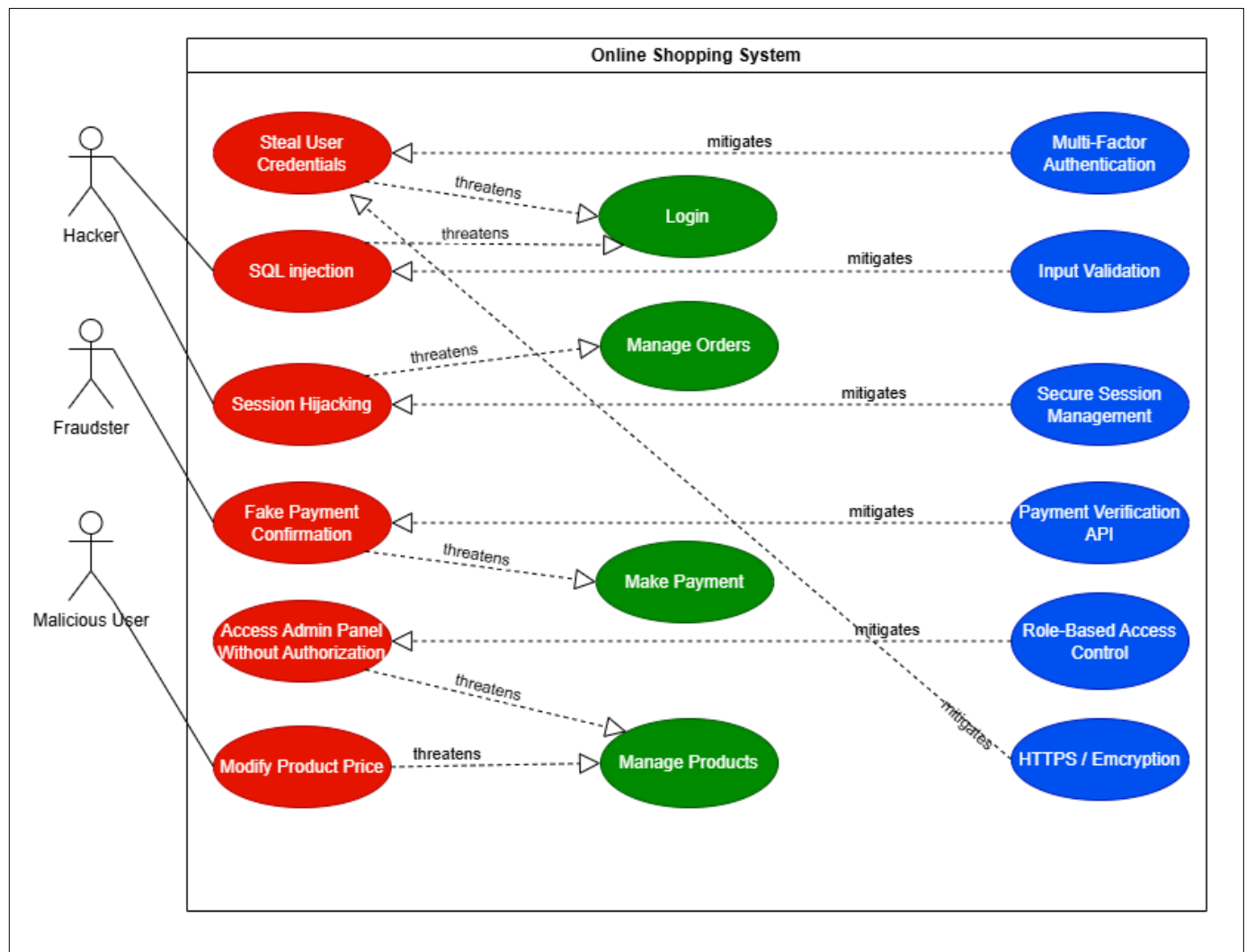
Unauthorized admin access, Role-based access control

Fake payment, Payment verification API

1. Use Case Diagram (UML)



2. Misuse Case Diagram



3. Table of Threats and Controls

Normal Use Case	Misuse (Threat)	Case	Threat Type	Security Control	Control Type
Login	SQL Injection		Injection Attack	Input Validation / Prepared Statements	Preventive
Login	Steal User Credentials		Credential Theft	Multi-Factor Authentication (MFA)	Preventive
Manage Orders	Session Hijacking		Session Attack	Secure Session Management	Preventive
Make Payment	Fake Payment Confirmation		Payment Fraud	Payment Verification API	Preventive
Manage Products	Access Admin Panel Without Authorization		Privilege Escalation	Role-Based Access Control (RBAC)	Preventive

Manage Products	Modify Product Price	Unauthorized Modification	Role-Based Access Control (RBAC)	Preventive
Login / Data Transfer	Credential Interception	Man-in-the-Middle Attack	HTTPS / Encryption	Preventive

This table clearly illustrates the relationship among:

Threat → Target → Control

4. How misuse case improves security

Boosting Security with Misuse Cases

Misuse cases play a critical role in enhancing system security by adding an attacker-centric lens to system modeling. While traditional use case modeling focuses on functional requirements—what the system should deliver to legitimate users—misuse case modeling broadens that view to identify how malicious actors might exploit vulnerabilities within the system.

By examining threats like SQL injection, credential theft, and unauthorized access, developers are prompted to evaluate potential weaknesses during the design phase instead of waiting until after deployment. This proactive stance aims to lessen security risks and avoid the costly fixes that often come with addressing issues post-implementation.

Misuse cases strengthen security through several essential mechanisms:

1. **Early Risk Spotting** – Identifying threats before any coding begins.
2. **Traceable Security Requirements** – Each threat is matched with a corresponding mitigation control.
3. **Heightened Threat Awareness** – Designers adopt an attacker’s perspective to foresee possible exploits.
4. **Thorough Security Documentation** – Security considerations become part of system specifications.
5. **Enhanced Access Control Design** – Encourages the principle of least privilege and robust authentication measures.

For example, identifying “SQL Injection” as a threat to the “Login” function leads to implementing input validation and parameterized queries. Without this foresight, such essential security controls could easily be overlooked.

Thus, misuse cases help shift the focus from a reactive defense strategy to a proactive design methodology.

5. Difference between use case and misuse case

Aspect	Use Case	Misuse Case
Purpose	Describes standard system functionality	Describes possible malicious actions
Actor Type	Legitimate users (Customer, Admin)	Malicious actors (Hacker, Fraudster)
Goal	Achieve business objectives	Exploit system weaknesses
Focus	Functional requirements	Security vulnerabilities
Relationships	«include», «extend»	threatens, mitigates
System View	Positive behavior	Negative/abusive behavior

In summary:

- A Use Case details the intended functionality of the system.
- A Misuse Case explains how attackers might leverage the system for malicious purposes.

Use cases address the question:

→ “How does the system create value?”

Misuse cases tackle the question:

→ “What potential attack vectors exist, and what can we do to defend against them?”

Together, they form a comprehensive system model that merges functionality with robust security resilience.

THE END
